

Tables (9)

Name	Type	Schema
Authors		CREATE TABLE Authors (author_id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT NOT NULL)
author_id	INTEGER	"author_id" INTEGER
name	TEXT	"name" TEXT NOT NULL
BookAuthors		CREATE TABLE BookAuthors (id INTEGER PRIMARY KEY AUTOINCREMENT, book_id INTEGER NOT NULL, author_id INTEGER NOT NULL, FOREIGN KEY (book_id) REFERENCES Books(book_id) ON DELETE CASCADE, FOREIGN KEY (author_id) REFERENCES Authors(author_id) ON DELETE CASCADE)
id	INTEGER	"id" INTEGER
book_id	INTEGER	"book_id" INTEGER NOT NULL
author_id	INTEGER	"author_id" INTEGER NOT NULL
Books		CREATE TABLE Books (book_id INTEGER PRIMARY KEY AUTOINCREMENT, title TEXT NOT NULL, genre TEXT NOT NULL, publisher TEXT, year INTEGER)
book_id	INTEGER	"book_id" INTEGER
title	TEXT	"title" TEXT NOT NULL
genre	TEXT	"genre" TEXT NOT NULL
publisher	TEXT	"publisher" TEXT
year	INTEGER	"year" INTEGER
Copies		CREATE TABLE Copies (copy_id INTEGER PRIMARY KEY AUTOINCREMENT, book_id INTEGER NOT NULL, status TEXT NOT NULL DEFAULT 'Available', FOREIGN KEY (book_id) REFERENCES Books(book_id) ON DELETE CASCADE)
copy_id	INTEGER	"copy_id" INTEGER
book_id	INTEGER	"book_id" INTEGER NOT NULL
status	TEXT	"status" TEXT NOT NULL DEFAULT 'Available'
Fines		CREATE TABLE Fines (fine_id INTEGER PRIMARY KEY AUTOINCREMENT, transaction_id INTEGER NOT NULL, amount REAL NOT NULL, status TEXT NOT NULL CHECK(status IN ('Paid','Unpaid')), FOREIGN KEY (transaction_id) REFERENCES Transactions(transaction_id))
fine_id	INTEGER	"fine_id" INTEGER
transaction_id	INTEGER	"transaction_id" INTEGER NOT NULL
amount	REAL	"amount" REAL NOT NULL
status	TEXT	"status" TEXT NOT NULL CHECK("status" IN ('Paid', 'Unpaid'))
Members		CREATE TABLE Members (member_id INTEGER PRIMARY KEY AUTOINCREMENT, name TEXT NOT NULL, email TEXT UNIQUE NOT NULL, phone TEXT, address TEXT)
member_id	INTEGER	"member_id" INTEGER
name	TEXT	"name" TEXT NOT NULL
email	TEXT	"email" TEXT NOT NULL UNIQUE
phone	TEXT	"phone" TEXT
address	TEXT	"address" TEXT
Reservations		CREATE TABLE Reservations (reservation_id INTEGER PRIMARY KEY AUTOINCREMENT, member_id INTEGER NOT NULL, book_id INTEGER NOT NULL, reservation_date DATE NOT NULL, FOREIGN KEY (member_id) REFERENCES Members(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id))
reservation_id	INTEGER	"reservation_id" INTEGER

Name	Type	Schema
member_id	INTEGER	"member_id" INTEGER NOT NULL
book_id	INTEGER	"book_id" INTEGER NOT NULL
reservation_date	DATE	"reservation_date" DATE NOT NULL
Transactions		CREATE TABLE Transactions (transaction_id INTEGER PRIMARY KEY AUTOINCREMENT, member_id INTEGER NOT NULL, copy_id INTEGER NOT NULL, issue_date DATE NOT NULL, return_date DATE, FOREIGN KEY (member_id) REFERENCES Members(member_id), FOREIGN KEY (copy_id) REFERENCES Copies(copy_id))
transaction_id	INTEGER	"transaction_id" INTEGER
member_id	INTEGER	"member_id" INTEGER NOT NULL
copy_id	INTEGER	"copy_id" INTEGER NOT NULL
issue_date	DATE	"issue_date" DATE NOT NULL
return_date	DATE	"return_date" DATE
sqlite_sequence		CREATE TABLE sqlite_sequence(name,seq)
name		"name"
seq		"seq"

Indices (2)

Name	Type	Schema
idx_book_title		CREATE INDEX idx_book_title ON Books(title)
title		"title"
idx_member_name		CREATE INDEX idx_member_name ON Members(name)
name		"name"

Views (1)

Name	Type	Schema
issued_books		CREATE VIEW issued_books AS SELECT m.name AS member_name, b.title AS book_title, t.issue_date FROM Transactions t JOIN Members m ON t.member_id = m.member_id JOIN Copies c ON t.copy_id = c.copy_id JOIN Books b ON c.book_id = b.book_id WHERE t.return_date IS NULL
member_name	TEXT	"member_name" TEXT
book_title	TEXT	"book_title" TEXT
issue_date	DATE	"issue_date" DATE

Triggers (2)

Name	Type	Schema
issue_book		CREATE TRIGGER issue_book AFTER INSERT ON Transactions BEGIN UPDATE Copies SET status = 'Issued' WHERE copy_id = NEW.copy_id; END
return_book		CREATE TRIGGER return_book AFTER UPDATE ON Transactions WHEN NEW.return_date IS NOT NULL BEGIN UPDATE Copies SET status = 'Available' WHERE copy_id = NEW.copy_id; END